

Solving $Ax = b$: Row Reduced Form R

Now the target isn't zero. When can it be hit, what is every portfolio that hits it, and why does the rank alone tell you whether there are none, one or infinitely many?

PART 1 Index

What this lesson covers, in order. Click a topic to jump to it.

1. Three possible outcomes

One answer, no answer, infinitely many: all three on 2×2 fund problems.

2. When is there a solution?

Eliminate with the target as letters; a zero row turns into a condition on the target: $R = 7B + 5E$.

3. The complete solution $x = x_p + x_n$

Free amounts to 0 for one answer, then add the special solutions. With a target tester.

4. Solving with R

$[A \mid b] \rightarrow [R \mid d]$: the particular solution is read straight off.

5. The rank decides everything

Four cases ($r = m = n$, $r = n$, $r = m$, r smaller than both), each with funds.

METHODS IN THIS LESSON

[Elimination on \$\[A \mid b\]\$](#)
[Solvability condition](#)
[Particular solution](#)
[Complete solution \$x_p + x_n\$](#)
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PART 2 Prerequisites

Click any link to review it, then come back.

- Special solutions and the reduced form R (Lesson 7).
- $Ax = b$ is solvable exactly when b is in $C(A)$ and one answer + the nullspace (Lesson 6).
- The plane $R = 7B + 5E$ (Lesson 5).
- Elimination with the right side (Lesson 2).

PART 3 Explanation

THE WHOLE IDEA IN ONE LINE

Eliminate $[A \mid b]$. If a row of A disappears but its b -entry doesn't, the target is **unreachable**. Otherwise set the free amounts to 0 to get **one** portfolio, then add every do-nothing trade to get **all** of them. The rank tells you in advance which of these can happen.

THE RUNNING EXAMPLE

The four-fund menu from Lesson 7: Nifty (1, 12, 1), Nifty ETF (1, 12, 1), bonds (1, 7, 0), balanced (1, 9.5, 0.5). Rules: budget B (₹ lakh), return R (₹ thousand), exposure E . A client asks for a target $b = (B, R, E)$:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 12 & 12 & 7 & 9.5 \\ 1 & 1 & 0 & 0.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} B \\ R \\ E \end{bmatrix}.$$

1. Three possible outcomes

THE WHOLE IDEA IN ONE LINE

A system $Ax = b$ has **exactly one** solution, **no** solution, or **infinitely many**. Never exactly two.

Three 2×2 problems with budget and return only show all three.

1 NIFTY + BONDS

$x + y = 10$, $12x + 7y = 105$. Row 2 $- 12 \times$ row 1: $0x + (7 - 12)y = 105 - 120$, so $-5y = -15$, $y = 3$, then $x = 10 - 3 = 7$.

OUTCOME

Exactly one: ₹7 lakh Nifty, ₹3 lakh bonds.

2 TWINS, ₹120K

Nifty and its ETF, target (10, 120). Row 2 $- 12 \times$ row 1: $0x + 0y = 120 - 12(10) = 0$. True, and it says nothing.

OUTCOME

Infinitely many: any split with $x + y = 10$, such as (10, 0), (4, 6), (-2, 12). All earn 12% on ₹10 lakh = ₹120 thousand.

3 TWINS, ₹105K

Target (10, 105). Row 2 $- 12 \times$ row 1: $0x + 0y = 105 - 120 = -15$. That says $0 = -15$.

OUTCOME

None. Two copies of a 12% fund can't earn 10.5%.

Why never two? If x and y both work, the trade $y - x$ is a do-nothing trade ($A(y - x) = b - b = 0$), and every multiple of it is too. So $x + t(y - x)$ works for every t : one extra solution forces infinitely many.

IN ONE LINE

A vanished row with a non-zero right side $\Rightarrow$ none; a vanished row with 0 and a free column $\Rightarrow$ infinitely many; no free column $\Rightarrow$ at most one.

2. When is there a solution?

THE WHOLE IDEA IN ONE LINE

Carry the target along as **letters** while you eliminate. When a row of A turns into zeros, its right side must be zero too. That gives the exact condition a target must meet.

METHOD · ELIMINATION ON THE AUGMENTED MATRIX $[A \mid B]$

Put b as an extra column and apply every row operation to it as well. Row operations don't change which x solve the system, so $[A \mid b]$, $[U \mid c]$ and $[R \mid d]$ all have the same solutions.

1 ROW 2

Row 2 $- 12 \times$ row 1, the A part as in Lesson 7: $(0, 0, -5, -2.5)$.
Right side: $R - 12B$.

ROW 3

Row 3 $-$ row 1: $(0, 0, -1, -0.5)$. Right side: $E - B$.

2 ROW 3 AGAIN

Row 3 $- 0.2 \times$ row 2: the A part becomes $(0, 0, 0, 0)$. Right side:
 $(E - B) - 0.2(R - 12B) = E - B - 0.2R + 2.4B = E + 1.4B - 0.2R$.

TIDY

Multiply by 5: $5E + 7B - R$. So the third row reads $0 = \frac{1}{5}(7B + 5E - R)$.

$$[U \mid c] = \left[\begin{array}{cccc|c} \boxed{1} & 1 & 1 & 1 & B \\ 0 & 0 & \boxed{-5} & -2.5 & R - 12B \\ 0 & 0 & 0 & 0 & \frac{1}{5}(7B + 5E - R) \end{array} \right]$$

RULE · THE SOLVABILITY CONDITION

$Ax = b$ is solvable exactly when every zero row of U has a zero right side. Here: $7B + 5E - R = 0$, that is $R = 7B + 5E$, the very plane $C(A)$ from Lesson 5. The algebra and the picture agree.

Target (B, R, E)	$7B + 5E - R$	Verdict
$(10, 105, 7)$	$70 + 35 - 105 = 0$	solvable ✓
$(10, 120, 10)$	$70 + 50 - 120 = 0$	solvable ✓ (all in Nifty)
$(10, 105, 5)$	$70 + 25 - 105 = -10$	no solution ✗

Why a row vanished. For these funds, exposure is a mix of the other two rules: $E = \frac{1}{5}(R - 7B)$. Whatever combination of rows killed row 3 of A must give 0 on the right as well. MIT's example says the same: row 3 = row 1 + row 2, so the target must have $b_3 = b_1 + b_2$.

IN ONE LINE

Each vanished row gives one condition on the target; meeting them all is exactly " b is in $C(A)$ ".

3. The complete solution $x = x_p + x_n$

THE WHOLE IDEA IN ONE LINE

Find **one** portfolio by setting the free amounts to zero. Then add any mix of the special solutions (the do-nothing trades from Lesson 7). That's all of them.

Step 1: one answer (the particular solution)

METHOD · PARTICULAR SOLUTION x_p

Set every free variable to 0 and back-substitute in $[U \mid c]$ for the pivot variables. (Any one solution would do; this is just the easiest.)

1 TARGET

FREE = 0

$b = (10, 105, 7)$, which passed the test. Right sides: $B = 10$, $R - 12B = 105 - 120 = -15$, last = 0 ✓.

ETF $x_2 = 0$, balanced $x_4 = 0$.

2 ROW 2

ROW 1

SO

$$-5x_3 - 2.5(0) = -15 \Rightarrow x_3 = 3.$$

$$x_1 + 0 + 3 + 0 = 10 \Rightarrow x_1 = 7.$$

$$x_p = (7, 0, 3, 0): ₹7 \text{ lakh Nifty, ₹3 lakh bonds.}$$

Step 2: add the nullspace

RULE · COMPLETE SOLUTION = PARTICULAR + NULLSPACE

$x = x_p + x_n$ with x_n any vector in $N(A)$: $A(x_p + x_n) = b + 0 = b$. The coefficient of x_p is always exactly 1 (MIT problem 8.1).

$$x = \begin{bmatrix} 7 \\ 0 \\ 3 \\ 0 \end{bmatrix} + c \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} -0.5 \\ 0 \\ -0.5 \\ 1 \end{bmatrix}$$

1 EXAMPLE

CHECK

PICTURE

$$c = 2, d = 2: (7 - 2 - 1, 0 + 2 + 0, 3 + 0 - 1, 0 + 0 + 2) = (4, 2, 2, 2).$$

Budget $4 + 2 + 2 + 2 = 10$ ✓. Return $48 + 24 + 14 + 19 = 105$ ✓.
Exposure $4 + 2 + 0 + 1 = 7$ ✓.

All solutions form a **plane** in $\mathbb{R}^4$, parallel to the nullspace plane but shifted to pass through x_p . Not a subspace: it misses 0.

Try it: any target, any free amounts

Budget B Return R Exposure E

ETF c Balanced d

$$\text{Test: } 7B + 5E - R = 7(10) + 5(7) - 105 = 0.$$

Particular (free = 0): row 2 gives $-5x_3 = R - 12B = -15$, so $x_3 = 3$; row 1 gives $x_1 = B - x_3 = 7$. So $x_p = (7, 0, 3, 0)$.

Complete: $x = x_p + 2s_1 + 2s_2 = (4, 2, 2, 2)$ (₹ lakh in Nifty, ETF, bonds, balanced).

Budget: $1(4) + 1(2) + 1(2) + 1(2) = 10 \checkmark$
 Return: $12(4) + 12(2) + 7(2) + 9.5(2) = 105 \checkmark$
 Exposure: $1(4) + 1(2) + 0(2) + 0.5(2) = 7 \checkmark$

Hits the target. Change c and d: it still does.

▼ For the curious: MIT's example in the same steps

$A = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix}$, $b = (1, 5, 6)$: condition $b_3 - b_2 - b_1 = 6 - 5 - 1 = 0 \checkmark$. With $x_2 = x_4 = 0$:
 $2x_3 = 5 - 2(1) = 3$, so $x_3 = \frac{3}{2}$; $x_1 + 2(\frac{3}{2}) = 1$, so $x_1 = -2$. Complete solution:
 $(-2, 0, \frac{3}{2}, 0) + c_1(-2, 1, 0, 0) + c_2(2, 0, -2, 1)$.

IN ONE LINE

One portfolio (free amounts = 0) + any do-nothing trade = every portfolio that hits the target.

4. Solving with R

THE WHOLE IDEA IN ONE LINE

Reduce all the way to $[R \mid d]$. Then the particular solution sits in the last column: pivot variables = d , free variables = 0.

METHOD · GAUSS-JORDAN TO $[R \mid D]$

From $[U \mid c]$: divide each pivot row by its pivot and clear above each pivot, doing the same to the last column. Each non-zero row of $[R \mid d]$ then reads "pivot variable + (free terms) = d_i ".

1 PIVOT 2 → 1

CLEAR ABOVE

Row 2 $\div (-5)$: $(0, 0, 1, 0.5 \mid -15 / -5 = 3)$.

Row 1 $-$ row 2: $(1 - 0, 1 - 0, 1 - 1, 1 - 0.5 \mid 10 - 3) = (1, 1, 0, 0.5 \mid 7)$.

$$[R \mid d] = \left[\begin{array}{cccc|c} \boxed{1} & 1 & 0 & 0.5 & 7 \\ 0 & 0 & \boxed{1} & 0.5 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \quad \begin{array}{l} x_1 + x_2 + 0.5x_4 = 7 \\ x_3 + 0.5x_4 = 3 \end{array}$$

Free amounts 0: $x_1 = 7$, $x_3 = 3$. That's $x_p = (7, 0, 3, 0)$ with no back substitution, and the special solutions come from R 's free columns exactly as in Lesson 7. One matrix gives both

halves of the answer.

▼ **Check yourself (MIT problem 8.2):** reduce $[U \mid c]$ with $U = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 4 \end{bmatrix}$, $c = (5, 8)$, and give the complete solution.

Row 2 $\div 4$: $(0, 0, 1 \mid 2)$. Row 1 $- 3 \times$ row 2: $(1, 2, 0 \mid 5 - 6) = (1, 2, 0 \mid -1)$. Free $x_2 = 0$: $x_p = (-1, 0, 2)$. Special ($x_2 = 1$): $(-2, 1, 0)$. Complete: $(-1, 0, 2) + c(-2, 1, 0)$. Check x_p in U : $-1 + 0 + 6 = 5 \checkmark$, $0 + 0 + 8 = 8 \checkmark$.

IN ONE LINE

$[R \mid d]$: pivot variables = d when free = 0; free columns give the do-nothing trades.

5. The rank decides everything

THE WHOLE IDEA IN ONE LINE

Compare the rank r with the number of rules m and funds n . $r = n$ means no free columns (never more than one answer). $r = m$ means no zero rows (never "no answer").

RULE · THE FOUR RANK CASES

An $m \times n$ matrix of rank r always has $r \leq m$ and $r \leq n$. Which equalities hold decides the number of solutions.

Case	R looks like	Solutions of $Ax = b$	With our funds
$r = m = n$ (square, invertible)	I	exactly 1, for every b	Nifty, bonds, midcap
$r = n < m$ (full column rank, tall)	$\begin{bmatrix} I \\ 0 \end{bmatrix}$	0 or 1	Nifty, bonds with 3 rules
$r = m < n$ (full row rank, wide)	$\begin{bmatrix} I & F \end{bmatrix}$	infinitely many, for every b	Nifty, bonds, midcap, balanced
$r < m, r < n$	$\begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$	0 or infinitely many	Nifty, ETF, bonds, balanced

Case 1: $r = m = n$: one answer, always

Nifty, bonds, midcap (3×3 , rank 3; Lesson 6). Target (10, 111, 7.4): the only portfolio is (5, 3, 2). Check: $5 + 3 + 2 = 10 \checkmark$; $60 + 21 + 30 = 111 \checkmark$; $5 + 0 + 2.4 = 7.4 \checkmark$. This is every lesson before this one.

Case 2: $r = n < m$: at most one

Only Nifty and bonds, but all three rules: $\begin{bmatrix} 1 & 1 \\ 12 & 7 \\ 1 & 0 \end{bmatrix}$, rank 2 (both columns pivot). No free column, so never two answers. But a zero row survives, so some targets fail: (10, 105, 7) gives (7, 3); (10, 105, 5) gives nothing (Lesson 5's unreachable target).

Case 3: $r = m < n$: infinitely many, always

Nifty, bonds, midcap, balanced: rank 3 = number of rules, so no zero row. Every target is reachable, in a line of ways ($n - r = 1$ free column).

1 TARGET

(10, 105, 5), impossible in case 2.

PARTICULAR

Balanced = 0, solve with Nifty, bonds, midcap: $x_p = (-1, 6, 5, 0)$. Check: budget $-1 + 6 + 5 = 10 \checkmark$; return $-12 + 42 + 75 = 105 \checkmark$; exposure $-1 + 0 + 6 = 5 \checkmark$.

COMPLETE

$(-1, 6, 5, 0) + t(-0.5, -0.5, 0, 1)$. It needs a short ₹1 lakh of Nifty at $t = 0$; some other t may suit the desk better.

Case 4: $r < m$ and $r < n$: none or infinitely many

The running example: rank 2, 3 rules, 4 funds. A zero row (so the test $R = 7B + 5E$ can fail) and free columns (so when it passes, there are infinitely many). (10, 105, 7): a plane of answers; (10, 105, 5): none.

READING THE CASES AS A TRADER

$r = m$: your products span every target you measure (a "complete" menu for those rules).

$r = n$: no product is redundant, so every answer is unique. Real desks usually live in case 3 or 4: more products than risks, and the spare freedom is used to pick the cheapest or most liquid route.

IN ONE LINE

$r = m \Rightarrow$ always solvable; $r = n \Rightarrow$ never more than one answer; both $\Rightarrow$ exactly one.

▼ **Check yourself (MIT recitation):** when does $x - 2y - 2z = b_1$, $2x - 5y - 4z = b_2$, $4x - 9y - 8z = b_3$ have a solution, and what is it?

Row 2 $- 2 \times$ row 1: $(0, -1, 0 \mid b_2 - 2b_1)$. Row 3 $- 4 \times$ row 1: $(0, -1, 0 \mid b_3 - 4b_1)$. Row 3 $-$ row 2: $(0, 0, 0 \mid b_3 - b_2 - 2b_1)$. So it's solvable exactly when $-2b_1 - b_2 + b_3 = 0$. Reducing: $y = 2b_1 - b_2$, $x = 5b_1 - 2b_2 + 2z$. Complete solution: $(5b_1 - 2b_2, 2b_1 - b_2, 0) + c(2, 0, 1)$. With $b = (1, 1, 3)$: $-2 - 1 + 3 = 0 \checkmark$, $x_p = (3, 1, 0)$.